

ALEXANDER DEAN

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PERSONAL SUMMARY

Systems and software engineer with a B.Sc. in Computer Science and 5.5+ years architecting, deploying, and maintaining production systems across government contracts in distributed Agile teams. Experienced with CI/CD (Azure DevOps, GitHub Actions, GitLab), infrastructure-as-code (Ansible, Terraform), and GitOps-driven Kubernetes delivery (ArgoCD, Helm, Traefik, cert-manager) across AWS and on-premises environments. Comfortable across the stack in Python, C# and TypeScript. Outside of work I run a six-cluster Kubernetes homelab on Proxmox and AWS with a full Grafana observability stack, GitOps via Github and ArgoCD, and self-hosted tooling exposed through a Pangolin/Newt tunnel mesh, the platform I use to access private and public personal projects such as my personal website and Sentinel (a Go-based SDLC orchestrator). This hands-on environment is where I sharpen the patterns I bring to production work.

TECHNICAL SKILLS

Infrastructure & DevOps: Terraform, Kubernetes, Helm, ArgoCD, Ansible, Git, Azure DevOps, AWS, Docker

Frameworks & Libraries: Angular, ASP.NET, Karma-Jasmine

Languages: Python, YAML, Bash/Shell, SQL, C#, TypeScript, VB.NET,

EXPERIENCE

Software Engineer, Senior Associate

Peraton – NAVAIR Theater Mission Planning Center

Aug. 2025 – Present

Hybrid – Blacksburg, VA

- Designed Ansible automation to install and configure Windows and Red Hat systems for the Navy's Tomahawk missile mission planning center, replacing legacy deployment workflows.
- Refactored existing scripts into idempotent Ansible roles and playbooks, enabling zero-touch deployment in Azure DevOps, significantly cutting estimated configuration and deployment time.

Software Engineer, Senior Associate

Peraton – DataOnyx Independent Research and Development

Jan. 2024 – Aug. 2025

Hybrid – Blacksburg, VA

- Helped to deploy and maintain project infrastructure using ArgoCD, Helm, and EKS on AWS to align with demonstration requirements.
- Built Retrieval-Augmented Generation (RAG) solutions over a MongoDB, Airflow, and Kafka stack in Python; presented results to leadership and investors to secure further project funding.

Software Engineer, Associate

Peraton – County of San Diego Integrated Property Tax System

June 2022 – Jan. 2024

Hybrid – Blacksburg, VA

- Delivered full-stack development for a County of San Diego property tax collections application on an Angular and .NET stack, working directly with the client.
- Integrated Salesforce REST APIs and maintained a Karma-Jasmine unit test suite with SonarQube quality gate enforcement to meet contract coverage requirements.

Software Development Intern

Mapcom Systems

Nov. 2019 – July 2021

Remote – Richmond, VA

- Wrote SQL queries to audit client database integrity and developed automated regression scripts for the Pinpoint Locates product as part of an Agile delivery team.

Software Development Intern

Paramont Manufacturing

May 2019 – Aug. 2019

Abingdon, VA

- Built and maintained VB.NET Windows applications integrating SQL Server, Active Directory, and Microsoft Office, improving data validity for production and freight departments.

EDUCATION

Virginia Tech

Graduate Coursework – Computer Science

Blacksburg, VA

Aug. 2021 – May 2022

University of Virginia's College at Wise

Bachelor of Science in Computer Science

Wise, VA

Aug. 2017 – May 2021